

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

✓ Inheritance Code Solution

```
class Vehicle{  
    private int regd;  
    private String m;  
  
    public void setm(String m) {  
        this.m = m;  
    }  
  
    // initialize regd  
    public void setregd(int regd) {  
        this.regd = regd;  
    }  
  
    // access m  
    public String getm() {  
        return this.m;  
    }  
  
    // access regd  
    public int getregd() {  
        return this.regd;  
    }  
}  
  
class Car extends Vehicle  
{
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

```
private String name;

private String color;

public Car(String name,String color)
{
    super();
    this.name=name;
    this.color=color;
}

public String getName()
{
    return this.name;
}

public String getColor()
{
    return this.color;
}
}
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

✓ Employee tax regime Solution

```
class Income
{
    public static void main(String args[])
    {
        double tax=0,it;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter income ");
        it=sc.nextDouble();
        if(it<=200000)
            tax=0;
        else if(it<=300000)
            tax=0.1*(it-200000);
        else if(it<=500000)
            tax=(0.2*(it-300000))+(0.1*100000);
        else if(it<=1000000)
            tax=(0.3*(it-500000))+(0.2*200000)+(0.1*100000);
        else
            tax=(0.4*(it-1000000))+(0.3*500000)+(0.2*200000)+(0.1*100000);
        System.out.println("Income tax amount is "+tax);
    }
}
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

✓ Students evaluation criteria code Solution

```
class Student extends Person {  
    constructor(firstName, lastName, id, scores) {  
        super(firstName, lastName, id, scores);  
  
        this.scores = scores;  
    }  
  
    calculate() {  
        const average = this.scores.reduce((target, score) => { return target + score }) /  
this.scores.length;  
  
        const range = (average < 70)  
            ? Math.ceil((100 - average) / 15)  
            : Math.floor((100 - average - 1) / 10);  
  
        let letter = "";  
  
        switch (range) {  
            case -1:  
            case 0:  
                letter = 'O';  
                break;  
  
            case 1:  
                letter = 'E';  
                break;  
        }  
    }  
}
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

case 2:

letter = 'A';

break;

case 3:

letter = 'P';

break;

case 4:

letter = 'D';

break;

default:

letter = 'T';

break;

}

return letter;

}

}

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

✓ Vehicle code Solution

```
class Vehicle{  
    private int regd;  
    private String m;  
  
    public void setm(String m) {  
        this.m = m;  
    }  
  
    // initialize regd  
    public void setregd(int regd) {  
        this.regd = regd;  
    }  
  
    // access m  
    public String getm() {  
        return this.m;  
    }  
  
    // access regd  
    public int getregd() {  
        return this.regd;  
    }  
}  
  
class Car extends Vehicle
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

```
{  
    private String name;  
    private String color;  
    public Car(String name,String color)  
    {  
        super();  
        this.name=name;  
        this.color=color;  
    }  
    public String getName()  
    {  
        return this.name;  
    }  
  
    public String getColor()  
    {  
        return this.color;  
    }  
}
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

✓ Custom sort object code Solution

// Java program to sort student

// data according to their marks

import java.util.*;

// implementing comparable interface

class StudentData implements Comparable<StudentData> {

String name;

int marks;

// Constructor

StudentData(String name, int marks)

{

this.name = name;

this.marks = marks;

}

// overriding method to sort

// the student data

public int compareTo(StudentData sd)

{

return this.marks - sd.marks;

}

}

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

```
// Driver class

class GFG {

    public static void main(String[] args)

    {

        ArrayList<StudentData> list

        = new ArrayList<StudentData>();

        // Inserting data

        list.add(new StudentData("Ram", 98));

        list.add(new StudentData("Shyam", 84));

        list.add(new StudentData("Lokesh", 90));

        Collections.sort(list);

        // Displaying data

        for (StudentData sd : list)

            System.out.println(sd.name + " " + sd.marks);

    }

}
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

✓ Real Estate Business code Solution

//Real Estate Business

```
boolean[][] matrix = new boolean[n][m];
```

```
for(int i = 0; i < n; i++) {
```

```
    for(int j = 0; j < m; j++) {
```

```
        if(clients[i][0] <= houses[j][0] && clients[i][1] >= houses[j][1] )
```

```
            matrix[i][j] = true;
```

```
    }
```

```
}
```

```
int [] matchR = new int [m];
```

```
Arrays.fill(matchR, -1);
```

```
int result = 0;
```

```
for(int u = 0 ; u < n; u++) {
```

```
    boolean[] isSeen = new boolean[m];
```

```
    if(findHouse(matrix, u, isSeen, matchR)) {
```

```
        result++;
```

```
    }
```

```
}
```

```
return result;
```

```
}
```

```
static boolean findHouse(boolean[][] matrix, int clientU, boolean[] isSeen, int[] matchR ) {
```

```
    int length = isSeen.length;
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

```
for(int v = 0; v < length; v++) {  
    if(matrix[clientU][v] == true && !isSeen[v]) {  
        isSeen[v] = true;  
        if(matchR[v] < 0 || findHouse(matrix, matchR[v], isSeen, matchR)) {  
            matchR[v] = clientU;  
            return true;  
        }  
    }  
}
```

```
return false;  
}
```

```
//custom sort
```

```
class Candidate implements Comparable<Candidate>{
```

```
    private String name;
```

```
    private int marks;
```

```
    public Candidate(String name,int marks)
```

```
{
```

```
        this.name=name;
```

```
        this.marks=marks;
```

```
}
```

```
    public int compareTo(Candidate candidate){
```

```
        if(this.marks<candidate.marks){
```

```
            return -1;
```

```
        }
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

```
if(this.marks>candidate.marks){  
    return 1;  
}  
return 0;  
}  
int getMarks()  
{  
    return this.marks;  
}  
String getName()  
{  
    return this.name;  
}
```

TECH LECTURE WITH MADHURI MOURYA

4/April/2021, 25/April/2021,26/April/2021

Apartment Garden Code Solutions

```
JAVA (1.8) v
5 import java . i . * ;
8 import java . useful . * ;
9 public class Main
10 {
11     public void solve(int[] heights){
12         for(int i=0;i<heights.length;i++)
13         {
14             for(int j=0;j<heights.length;j++)
15             {
16                 if(i==j-1 && heights[i]>heights[j])
17                 {
18                     System.out.println(i);
19                 }
20             }
21         }
22     }
23     public static void main(String[] args){
24         int n ;
25         Scanner sc=new Scanner(System.in);
26         n=sc.nextInt();
27         int[] array=new int[n];
28         for ( int i = 0 ; i < n ; i ++ )
29         {
30             array[i]=sc.nextInt();
31         }
32         sc.close();
33         Main obj = new Main();
34         obj.solve(array);
35     }
36 }
```

Tech Lecture with Madhuri